

Process of ID

1.

What is the current user experience (UX)

(19) The current user experience is very poor for everyone involved. Subsistence farmers often produce surplus crops. However, they cannot sell or donate their excess produce because they live in rural areas with poor transport infrastructure. As a result, the surplus produce goes to waste causing farmers to feel guilty and lose potential income. At the same time, local traders experience uncertainty and inconvenience when sourcing fresh produce. They often find it difficult to identify nearby farmers who have surplus produce available, which then forces them to rely on expensive suppliers. Many households experience food insecurity even when there is plenty of food available nearby, they have no way of knowing when and where food is available. This disconnect creates an inefficient and frustrating experience for everyone involved.

Their UX?

-2

Other, explicitly mentioned, UX?

What usability goals are currently met or not met

None of the usability goals are met because there is no system that connects farmers, traders, transport providers and households in need. The system is **ineffective** because users cannot easily sell, buy, donate or transport surplus produce. It is also **inefficient** because users have to spend more time and effort to achieve their goals. There is no central platform that connects the different stakeholders, so every step of the process is unnecessarily difficult. Farmers must try to find buyers or recipients on their own or rely on verbal communication to get information to the right ears. **Utility** is also not met because farmers cannot advertise their surplus produce, traders cannot easily search for nearby suppliers, transport providers cannot view delivery requests and households cannot find available food. Furthermore, users experience low **satisfaction** because food is wasted while some families remain hungry

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Who are the stakeholders and specific users of the proposed system

The primary users of the proposed system include small scale farmers, traders, vulnerable households and transport providers. Small scale farmers would use the system to list their

surplus produce. Traders such as markets, hawkers and spaza shop owners would purchase the fresh produce listed on the system. Households could use the system to access donated or reasonably priced food. Transport providers would help to help to transport the produce from farmers to traders or households.

System developers are also important stakeholders because they are responsible for designing, developing, testing and maintaining the platform to ensure that it meets users' needs and functions reliably. Other stakeholders include non-governmental organisations (NGOs) who aim to support small farmers and improve food insecurity. ✓

How and why would your proposed solution address the identified challenge (you could introduce the type of solution you have in mind here)

The solution to address the challenge of surplus crops, poor transport infrastructure and food insecurity in many households is to introduce the radio broadcast system. Steps to introduce the system starts through community meetings that are led by local leaders to inform the rural farmers about the new radio system that will allow them to announce their surplus crops and also explain to them how it works, its purpose, and how the farmers can call or send messages to the radio station.

Then the design team collaborates with a community radio station that already broadcasts the rural areas to introduce a slot that can be called a farmers marketing time which can be scheduled at a specific time daily so that farmers and traders know when to tune in.

The farmers can use their phones to call in the station directly or SMS and the farmers who do not have phones can use community shared call points like at clinics or community halls to call the station and announce information about their products surplus and also leave their contact information. The radio hosts can also invite farmers to the station or let them call in live during the farmer's slot so that the farmers can be reassured that their voices are heard or they receive confirmation messages to confirm that their announcement was broadcasted. This can also build the farmer's confidence.

Local traders and transport providers can respond by contacting the farmers directly or through to request the products and keep the broadcast relevant.

The reason we propose the radio broadcast as the solution is to make it more accessible to rural farmers in the Eastern Cape as they may not be exposed to advanced technology like smartphones or online systems. Radio is familiar and it is used the most by rural areas, transport providers in their cars tune in every time to listen, most households and spaza

Would this be, e.g., a phone at such a point? Would this be efficient?

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shops have it. They don't need to learn new tools, they just call to announce their surplus produce on radio channels like Vukani community Radio. These Radio broadcasts will make the surplus visible to transporters or households. Radio also saves costs as it requires minimal investment. Farmers know when to announce and households know when to tune in which build trust and assure them that the system is reliable. This solution will relieve all the frustration, and it will empower farmers to share their surplus, earn income and reduce wastes while households get to access food affordably.

Which two design principles would be most appropriate to consider for your solution

Visibility: The Radio broadcast is at a fixed and predictable slot, so all the stakeholders know when to tune in and listen to the announcements and the stations repeat announcements to make sure that they are heard. The information announced contains all the details like product type and quantity, so listeners are able to understand quickly. This ensures that all the stakeholders can clearly see or hear the flow of information and access the surplus information, regardless of their technological background.

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Not on a radio broadcast. Visibility is about exactly that - making things visible.

Consistency: Radio broadcast is what they are familiar with, they don't need to learn about new tools. Farmers and traders should know that announcements occur regularly at a specific time so they can rely on the system which builds confidence among farmers and households. It ensures that the radio broadcast system becomes a part of daily life.

2.

- a) Lesson three: What it really takes to scale Design Solutions, primarily focusing on the principle of "**Designing for Implementers**".
- b) "Designing for implementers" stipulates that people who will be operating the solution that is devised hold critical knowledge that designers may not possess. Consequently, they must be treated as end-users of the system as well, rather than being consulted afterwards.

(5)

I would identify everyone who will implement the system, such as farmers, hawkers, and vulnerable households.

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Rather than asking these stakeholders what they think is necessary for the system to meet their needs, I would spend time observing their everyday routines. This would reveal realities that already exist, such as trusted networks and other informal systems already in place.

Where trust doesn't already exist, for example, between a farmer and a household who don't know each other, I would look to existing trusted intermediaries in the community, such as spaza shops, etc, to broker **initial introductions**, since stakeholders are more likely to engage through someone they already trust than an unfamiliar new system.



By centering implementers early, I ensure that the eventual design properly fits into the current networks and how stakeholders are operating, instead of forcing them to adopt a new design that might break and collapse after being introduced to real-world logistics, relationships, and constraints. [197]